<u>Detailed and Elaboratory Thesis chapter breakdown</u>

1.1 Chapter Overview

This chapter covers the research topic "A Weather Based Vegetable Price and Crop Decision Support Management Information System for Sri Lanka." It starts by discussing the real-world challenges smallholder farmers face. The problem statement includes both general and specific elements, supported by recent data. The chapter also presents the main research question, motivation, aim, specific objectives, a broad view of the proposed solution, resource needs, project scope, and concludes with a summary. This sets a solid foundation for the entire study.

1.2 Problem Background

Sri Lanka's agriculture sector is a key part of the economy. It employs many people and is vital for food security. Smallholder farmers mainly carry out vegetable cultivation on small plots of land. These farmers are very sensitive to changes in the market.

Recently, vegetable prices have experienced extreme ups and downs. For instance, during 2024 and 2025, the prices of vegetables like carrots, beans, and onions spiked after heavy rainfall and droughts damaged crops. When supply suddenly decreases, prices rise, causing problems for consumers. Conversely, when there is excess production, prices drop, leading to losses for farmers.

Government organizations, including the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Department of Agriculture, publish daily price reports. Some mobile apps, such as Govi Mithuru and GeoGoviya, provide current price and weather information. However, most of these tools only reflect current or past situations. Farmers still struggle to predict future trends, especially in uncertain weather. This issue has persisted for many years and worsened due to climate change.

1.3 Problem Statement

1.3.1 General Problem

The underlying issue is that frequent weather-related uncertainties in vegetable production lead to significant price fluctuations nationwide. This creates serious consequences, such as unstable incomes for farmers, higher food prices for consumers, and challenges in maintaining a steady food supply. Ultimately, it impacts the economic stability of the agriculture sector and contributes to food insecurity in Sri Lanka.

1.3.2 Specific Problem

In agricultural information systems and decision support, there are clear gaps. While some studies have used machine learning to predict prices, few practical systems have been developed that integrate real weather data with price forecasting. These systems typically offer limited dashboards. Most existing tools only display current prices and basic weather updates. They do not aid smallholder farmers in making proactive decisions about which crops to grow or when to sell.

This research gap needs attention since smallholder farmers, who make up most of the vegetable producers in Sri Lanka, still rely on guesswork or outdated information. A dedicated weather-based Management Information System with a straightforward dashboard can help fill this gap and offer timely, actionable support.

1.4 Research Question

How can a weather-based Management Information System with an interactive dashboard be designed and developed to assist smallholder farmers in Sri Lanka with short-term vegetable price prediction and crop decision-making?

Sub-questions include:

- Which weather and market factors most strongly affect vegetable prices?

- How useful will the dashboard be for farmers with limited technical skills?

1.5 Research Motivation

I chose this research topic because of the real challenges faced by smallholder farmers in my country. During field visits and informal talks with farmers in various areas, I witnessed how they lose money when prices suddenly drop due to bad weather. Many expressed a need for better information to help them plan ahead rather than react after issues arise.

As a final-year student in Management Information Systems, I wanted to work on a project that links technology to a real social need. This research aligns with national interests in using digital tools for agriculture and increasing climate resilience. I believe this system can make a meaningful impact on the lives of small farmers.

1.6 Research Aim

The goal of this research is to design, develop, and evaluate a weather-based Vegetable Price and Crop Decision Support Management Information System featuring a user-friendly Power BI dashboard that helps smallholder farmers in Sri Lanka make more informed decisions.

1.7 Research Objectives

1.7.1 To identify the key weather parameters and market factors that significantly influence the prices of selected vegetables (carrots, beans, leeks, onions, and pumpkins) in Sri Lanka.

1.7.2 To analyze historical vegetable price data and corresponding weather records to discover patterns and relationships.

1.7.3 To design and develop a web-based Management Information System with an interactive Power BI dashboard that offers price predictions and crop recommendations.

1.7.4 To evaluate the accuracy of the prediction models and the usability of the dashboard through testing with smallholder farmers and relevant stakeholders.

1.8 Rich Picture of the Proposed Solution

The proposed system follows a clear workflow. First, data is collected from two main sources: weather information from the Department of Meteorology and historical price data from HARTI and public datasets. This data is then cleaned and used to train machine learning models (such as Random Forest or time-series models). The models generate short-term price predictions.

Finally, all results are shown on a Power BI dashboard. Farmers and extension officers can access the dashboard to view price forecasts, trend charts, regional maps, and simple crop recommendations. The system emphasizes easy visualization so that even users with basic computer skills can understand the information and make timely decisions.

1.9 Resource Requirements

1.9.1 Hardware

Development can occur on a standard personal laptop with at least 8 GB RAM and a modern processor. For testing and sharing the dashboard, free or low-cost cloud services can be used.

1.9.2 Software

- Python programming language with libraries such as Pandas, Scikit-learn, and Matplotlib for data analysis and model building.
- Power BI Desktop for designing the interactive dashboard.
- Microsoft Excel or MySQL for initial data management.
- Publicly available datasets from Kaggle and official government sources.

1.10 Project Scope

Aspect	In Scope	Out of Scope
System Type	Web-based interactive Power BI dashboard	Mobile application development
Vegetables Covered	Carrots, beans, leeks, onions, pumpkins	Other crops and fruits
Main Features	Price prediction, visualizations, basic crop recommendations	Real-time IoT integration, automated buying/selling
Target Users	Smallholder farmers and agricultural extension officers	Large commercial farms and export companies
Evaluation	Model accuracy and basic usability testing with 5-10 farmers	Large-scale nationwide deployment and long-term impact study

The scope is kept manageable for an undergraduate project while still addressing the core problem.

1.11 Chapter Summary

This chapter provided a clear overview of the research. It discussed the current situation of vegetable price volatility in Sri Lanka, defined the problem, stated the research question, and outlined the aim and specific objectives. The proposed dashboard solution and project boundaries were also described. The next chapters will present the literature review, methodology, and other aspects of the study in detail.

Chapter – 02 [Literature Review]

2.1 Chapter Overview

The current chapter is a complete literature review about the creation of weather-based price prediction and decision support management information system of vegetables in Sri Lanka. The literature review is organized in a way to provide a reader with an overview of the field under consideration, then to analyze and criticize existing systems and frameworks, further to perform technological analysis of the algorithms, design, and workflow. After that, personal reflections will be provided as the evidence of the research gap. In general, the majority of literature sources selected were written within the last six years (2020-2025).

2.2 Conceptual Map of the Literature

For the sake of making navigation easy for the reader, the literature is divided into four major parts. The part 2. 3 provides an overview of the domain on the surface level but very important one. The part 2.4 discusses a comparative analysis of the existing systems and frameworks along with the personal assessment of their appropriateness. The part 2.5 provides a detailed technological analysis which includes algorithmic, design and workflow-related discussion. The part 2.6 provides reflections and convincingly demonstrates why there exists such a research gap based on recent literature findings.

2.3 Domain Overview

The agriculture industry of Sri Lanka remains a vital pillar of the economy of the country, contributing approximately 7–8% to the Gross Domestic Product (GDP) and providing employment to nearly one-quarter of the labor force. [1], [2] This sector makes an important contribution to the Gross Domestic Product, offers employment to many citizens, and plays a significant role in guaranteeing food security in the country. Vegetable farming is of particular importance in the agricultural industry because it supplies necessary produce for the citizens and generates regular income for rural families in the country.

At present, the process of vegetable production in Sri Lanka is characterized by the dominance of smallholder farmers cultivating on small pieces of land – usually less than two hectares. [3] Farmers have limited financial resources and are highly exposed to various threats. One of the most crucial problems is high price volatility associated with changing weather conditions. Weather fluctuations have a negative impact on crop development at all stages of its growth and may result in decreased productivity and even crop losses.

The recent effects of climate change have further increased weather unpredictability in Sri Lanka. Long periods of drought followed by heavy rainfall have become more common. Consequently, the availability of vegetables in the market fluctuates significantly, resulting in extreme price changes [4]This can be evidenced by the significant increase in the prices of some vegetables such as carrot, bean, leek, onion, and pumpkin during 2024 and 2025 in important economic hubs like Dambulla and Colombo due to unfavorable weather conditions [5]. Prices of these products increase whenever there is an imbalance in supply. However, when favorable weather results in high production levels, prices fall drastically, leading to great losses for farmers.

Some government bodies such as Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Department of Agriculture have taken steps to help farmers through the publication of price reports on a daily or weekly basis in websites and mobile applications. There are apps such as Govi Mithuru and GeoGoviya which offer prices in the market along with the weather forecast. Although these resources are very useful, most of them only offer historical or present information without the forecast.

The review of the above domain has revealed that price instability caused by weather is a persistent and escalating issue within the vegetable industry in Sri Lanka. This impacts both the economy of the smallholder farmers as well as increasing food prices, which ultimately affects the country's food security. Hence, there is a good reason for conducting research in this area.

2.4 Existing Systems / Frameworks / Designs

There have been multiple applications and frameworks created both in the local and international scenario to solve problems regarding agricultural decisions. Some of the popular mobile applications used in the Sri Lankan case are Govi Mithuru (created by Dialog Axiata), GeoGoviya, Krushi Advisor, and Ruhuna Govi-Nena. These applications offer daily vegetable prices, weather forecast, and some general cultivation tips. [6], [7], [8], [9].The good thing about these systems is that they provide much more information than before for many farmers and they have been quite successful in raising awareness. But after analyzing their features and after discussing with some farmers in the initial stage, I believe there are quite a few problems with these applications. They only give current or previous prices and general weather information. They cannot predict any future prices based on weather information. And for small scale farmers who have to make the decisions on which crops to grow for several weeks in advance, these applications are only reactive and not proactive enough.

Academically, there are few local research works done about using machine learning algorithms to predict vegetable prices. Notably, there is a study conducted in 2023 in the journal of management matters where they have used algorithms like tree-based random forest. They have used rainfall, temperature, and production factors and concluded that tree-based algorithms were better at predicting prices of up-country and low-country vegetables [6] Moreover, another 2023 research work has been done regarding the market data of vegetables like beans and carrots in the Dambulla market. They have compared various algorithms and concluded that the performance of tree-based algorithms was better than others in terms of Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) [10]Prior to that in 2021, there was a research done using neural network and classifiers on the prices of beans, carrots, and pumpkin. Red onions in the Jaffna peninsula have also considered weather variables in their prediction [11]

These works are very relevant to the present research as they prove that combining weather information into the model could increase the accuracy of predictions considerably. Personally, I support the results presented by the authors because my preliminary research of the price data on Kaggle confirmed the correlation of prices and weather conditions. Nevertheless, the problem with these works is that they only reached the stage of modeling and almost none of them tried to develop an end-user tool that would be available to the farmers.

On an international scale, there are more holistic frameworks such as Climate Smart Agriculture Decision Support Systems and multiple Farm Management Information Systems (FMIS). Some of these frameworks have the integration of IoT sensors, satellites and analytics. Although very technically advanced, these systems are generally built for large-scale commercial farming in well-developed countries and require expensive equipment, high-speed internet access and highly-technical expertise – something which is not available easily for Sri Lankan smallholder farmers. Therefore, despite their usefulness, international frameworks cannot be used without extensive adaptation.

In conclusion, literature review indicates a lot of evidence regarding the technical feasibility of price prediction based on weather data. Nonetheless, there is still a lack of practical frameworks that can be utilized by Sri Lankan smallholder vegetable farmers

2.5 Technological Analysis

2.5.1 Algorithmic Analysis

Numerous algorithms have been employed in price prediction in agriculture. Classical approaches like ARIMA and Prophet are capable of modeling seasonal trends within historical prices. Moreover, machine learning algorithms including Random Forest, Gradient Boosting Machines and Support Vector Regression demonstrated great accuracy when weather factors (rainfall, temperature, humidity) were included to the set of features [6], [10]

As I understand, the hybrid approach which combines Random Forest for its interpretability and stability with LSTM for processing time-series is the most appropriate method in this study. The reason is recent research conducted in Sri Lanka which has proven the efficiency of tree-based algorithms for working with local datasets with missing values and nonlinearities. Moreover, LSTM will perform better in identifying the seasonality of vegetable prices due to monsoons. These algorithms are feasible to implement on a regular computer since they do not require expensive GPUs.

2.5.2 Design Analysis

A number of innovations have taken place in the development of information systems for agriculture. Early information systems relied solely on SMS alerts and simple mobile apps, whereas contemporary systems rely on interactive dashboards which are built using software like Power BI, Tableau, or web technologies. Interactive dashboards give several benefits including visualization (trends in prices and heat maps per region), and updates which are available on central computing devices from agrarian service centers [12] Power BI was selected as the software for this study because of the following features: interactivity, ease of learning without knowing any programming language, update through central server, accessible through both the web browser and the desktop, and low cost.

In this context, a Power BI dashboard design is the best design option for our project due to its support for good data visualizations, ease of learning and non-programming based nature of maintenance. Simplicity needs to be the key factor while designing the system with clear font size, color coding and minimum jargon. The justification of this design approach comes from the fact that existing mobile applications are very popular but suffer from the issue of being overwhelming with information to the user.

2.5.3 Workflow Analysis

The typical process flow of the present literature is consistent with the traditional data science process flow: data acquisition from various sources, data pre-processing (data cleaning, feature engineering, and data normalization), model training and testing, predictions, and lastly results presentation to the user [10] The new system would follow a similar process flow but with some improvements. Weather data from the Department of Meteorology and price data from

HARTI/Kaggle would be obtained regularly. Post pre-processing, models would be trained. Predictions and crop recommendations would then be presented on the Power BI dashboard.

The process flow is reasonable and directly follows from recent approaches. The implementation of a good visualization stage at the end of the process helps to overcome a shortcoming of the previous approaches, where the output was technically presented. The selected process flow is feasible and appropriate for the Sri Lankan context.

2.6 Reflection

Based on a critical analysis of recent literature from 2020 to 2025, a number of key findings can be outlined. First of all, machine learning algorithms together with weather data can predict prices of vegetables accurately [6] Still, there is a gap when it comes to creating usable tools that could be used by smallholder farmers on their daily basis. While all articles focus on increasing the accuracy of models, the usability of the developed tools and their ability to be adopted by farmers who have no technical knowledge are largely neglected.

In my opinion, a good prediction model is one side of the problem, and the other part is to present it in an easy-to-use manner. In order to bridge this gap, I have chosen the research topic related to developing a Power BI dashboard which will include the forecast of the weather, price prediction, and recommendations regarding crops to grow. This research will critically link the findings of the reviewed literature with its weaknesses and justify the necessity of the proposed system.

Chapter – 03 [Methodology]

3.1 Research Paradigm

The current research is guided by the Paradigm of Pragmatism. It is necessary to mention that pragmatism has been selected as the paradigm because the primary aim is to develop a solution for a pragmatic problem - smallholders' need for better decision making. Pragmatism differs from positivism (focused purely on objective measurements) and interpretivism (focused on subjective interpretations) in that it enables the researcher to consider both quantitative data (prediction models of price) and qualitative aspects (feedback of farmers). The pragmatic approach allows one to provide a clear perspective - the issue is studied from a pragmatic standpoint, meaning that the quality of the solution depends on its practical significance.

3.2 Research Approach

A mixed-method approach (pragmatic) is used, combining deductive and inductive elements. The deductive part involves testing existing machine learning algorithms on Sri Lankan data. The inductive part comes from observing farmer needs through surveys and interviews, then building the system based on those real needs. This combined approach is suitable because the research needs both technical model building and understanding of user requirements.

3.3 Research Strategy

This research strategy is Design Science Research (DSR). The use of DSR method is justified by the very essence of the output that will be created during this project the artifact that needs to be developed and evaluated is the Management Information System based on the Power BI dashboard.

3.4 Fact Collection Mechanisms

Information collection is done through secondary as well as primary modes.

Secondary information:

Price history of vegetables from Kaggle data sets as well as HARTI, and meteorological information (rainfall and temperature data) from the Department of Meteorology.

Primary information:

Survey and short interviews conducted among 5-10 smallholder farmers regarding their requirements and issues concerning the dashboard prototype.

This way we get a combination of both historical as well as firsthand information.

3.5 Research Methodology Execution Workflow

The execution follows Peffers' Design Science Research Methodology (DSRM). The following table explains how each stage is addressed in this research:

Stage	Description in this Research
3.5.1 Problem Identification	Discovered through literature reviews, discussions with farmers, and observations on price fluctuations in Sri Lankan vegetable industry.
3.5.2 Relevance Justification	The problem is relevant due to climate change impacts and the lack of predictive tools for smallholder farmers.

3.5.3 Comparative Analysis and Gap Justification	Compared other applications and researches; identified gap related to the absence of a simple dashboard based predictive application.
3.5.4 Define and Finalize Objectives 3.5.5 Design / Development / Data Management	Objectives were defined after the literature study and initial farmers' feedback
3.5.5 Design / Development / Data Management	Collected data and processed it; built ML models; designed Power BI dashboard.
3.5.6 Evaluation and Communication	The models were evaluated by means of MAE, RMSE, the dashboard was tested by farmers through a questionnaire on usability.

3.6 Project Management Methodology

Project Management Technique Agile (Scrum) is applied. This methodology is applicable since the task at hand involves an implementation project with requirements that will evolve from the responses of the farmers. Iterative development (with sprints lasting 2 – 3 weeks) is achievable through scrum. The Waterfall approach, among other traditional approaches, is not applicable since user response is critical during development.

3.6.1 Project Timeline

This will run from 02 December 2025 through 21 September 2026. Critical stages will be: review and collection of literature/data (first three months), development of system design (next five months), evaluation and testing (three months), and reporting (the remaining period of time).

3.6.2 Ethical Considerations   02 December 2025-21 September 2026

Ethical clearance shall be sought from the university. Informed consent shall be sought from all farmers who have participated. The data will be made anonymous and used for research purposes alone. The privacy of farmer information is carefully protected. The aim of this study is to help farmers without hurting anyone.

3.7 Chapter Summary

In this chapter, the research paradigm, approach, strategy, data collection methods, implementation process, and project management process have been outlined. Pragmatism and Design Science Research were explained to be the most appropriate paradigms for this development project. Future chapters will describe the design, implementation, and evaluation results of the system.

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